

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-607: Probability

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. All questions are of equal value.*

Question 1.

- (a) An experiment consists of tossing a coin until a head appears. Find its sample space. If the coin is fair, what is the probability that the coin will be tossed at most 10 times? [3]
- (b) Two fair dice are tossed. Find the probability that:
- (i) the sum is at least 3;
 - (ii) the sum is at most 10.

[3]

- (c) If X is a random variable with pdf:

$$f(x) = \begin{cases} C(1 - x^2), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

find the value of C .

[2]

- (d) If X is uniformly distributed over $(-1, 1)$, find the pdf of X and calculate $E[X^2]$. [2]
- (e) If the moment generating function of a random variable X is $M_X(t) = \frac{e^t}{2 - e^t}$, find the probability mass function of X and calculate $P(X = 1)$. [3]
- (f) If $X_1, X_2, \dots, X_{10}$ are independent binomial random variables each with mean 1, use Markov's inequality to find a bound for $P(X_1 + X_2 + \dots + X_{10} \geq 15)$. [2]
- (g) Suppose $p(x, y)$, the joint pmf of X and Y , is given by $p(1, 1) = 0.5$, $p(1, 2) = 0.1$, $p(2, 1) = 0.1$, $p(2, 2) = 0.3$. Find the conditional pmf of X given $Y = 2$. [3]

Question 2.

- (a) Sixty percent of families own a car, thirty percent own a house, and twenty percent own both. If a family is randomly chosen, find the probability that the family owns a car or a house but not both. Assuming the chosen family owns a car, what is the probability it does not own a house? [7]

- (b) Define the Poisson random variable. Prove that the Poisson distribution is the limiting case of the binomial distribution. [7]
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Question 3.

- (a) Define the expectation of a random variable. Calculate $E[X]$ and $E[X^2]$ when X is normally distributed with parameters μ and σ^2 . [7]
- (b) Define jointly distributed random variables. If X and Y are jointly distributed and $a, b \in \mathbb{R}$, prove that $E[aX + bY] = aE[X] + bE[Y]$. [7]
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Question 4.

- (a) Define and give an example of independent random variables. If X and Y are independent Poisson random variables with parameters λ_1 and λ_2 , find the probability distribution of $X + Y$. [7]
- (b) Define the moment generating function of a random variable. Calculate the mgf of an exponential random variable X . Hence find the mean and variance of X . [7]
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Question 5.

- (a) State and prove Chebyshev's inequality. Suppose the weekly production in a factory is a random variable with mean 500 and variance 100. What can be said about the probability that this week's production is between 400 and 600 items? [7]
- (b) Define conditional expectation. If the joint density of X and Y is:

$$f(x, y) = \begin{cases} 6xy(2 - x - y), & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

find the conditional expectation of X given $Y = y$ for $0 < y < 1$. [7]

Question 6.

- (a) Define conditional variance. If X and Y are two random variables, prove that:

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$$

[7]

- (b) Define probability by conditioning. n people have been invited to a party beginning at time 0. Arrival times of n guests are independent exponential random variables with mean 2, and the host arrives at a time uniformly distributed on $[0, 2]$.
- (i) Find the probability that exactly k guests arrive before the host.
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- (ii) Find the expected number of guests who arrive before the host.

[7]

Question 7.

- (a) In an election, candidate A receives m votes and candidate B receives n votes ($m > n$). Assuming all orderings are equally likely, show that the probability that A is always ahead of B in the vote count is $\frac{m-n}{m+n}$. [7]
- (b) Define the entropy of a system. Prove that the entropy is maximum if all events are equally probable. When is the entropy minimum? Explain. [7]